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Trends and advances in cardiac rehabilitation for congenital heart disease: a bibliometric analysis

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Abstract

Background Congenital heart disease (CHD) survivors often suffer from impaired health-related quality of life (HRQoL), requiring cardiac rehabilitation (CR) as an additional intervention. CR involves various modalities and has been demonstrated to reduce mortality and improve HRQoL in cardiovascular diseases. However, there is a lack of systematic analysis and summary of the accumulated literature on cardiac rehabilitation for congenital heart disease (CRfCHD). This study aimed to provide a comprehensive overview of the literature on cardiac rehabilitation for congenital heart disease. By analyzing research trends and hotspots, it also sought to highlight key developments and suggest future directions in this growing field.

Methods A systematic search of the Web of Science Core Collection was performed to identify publications related to CRfCHD. Bibliometric tools, including CiteSpace and VOSviewer, were used to visualize co-authorship networks, keyword clusters, and citation patterns.

Results A total of 218 articles were included in the analysis. The number of research papers on CRfCHD has shown an increasing trend over time, with a significant rise in publications since 2019. The United States has the highest number of publications, while the most cited paper focuses on the effects of physical exercise training in CHD patients. Keyword analysis reveals hotspots in the field such as congenital heart disease, physical activity, children, and adolescents.

Conclusion This bibliometric study analyzed the development of CRfCHD from 2003 to 2022, and overviewed the evolution and frontiers in this promising field. Future research should focus on developing comprehensive and effective CRfCHD programs. Home-based cardiac rehabilitation and mental health research in CHD patients require more attention.

Keywords Cardiac rehabilitation, Bibliometrics, Family practice, Quality of life, Physical activity, Mental health

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Introduction

According to the 2018 AHA/ACC Guideline for the Management of Adults With Congenital Heart Disease (CHD), CHD refers to structural abnormalities of the heart or great vessels that are present at birth, with a wide range of anatomical complexity and physiological consequences. These defects include atrial septal defects, characterized by an abnormal opening between the atria that permits left-to-right blood flow; ventricular septal defects, involving a hole in the ventricular septum allowing blood to pass between the ventricles; and patent ductus arteriosus, where a fetal vessel connecting the aorta and pulmonary artery fails to close after birth. More complex anomalies, such as tetralogy of Fallot—a combination of four cardiac abnormalities—and transposition of the great arteries, in which the aorta and pulmonary artery are transposed, are also classified as CHDs. Additional examples include coarctation of the aorta, which is a narrowing that restricts systemic blood flow; truncus arteriosus, where a single arterial trunk arises from the heart instead of separate pulmonary and aortic vessels; and hypoplastic left heart syndrome, marked by severe underdevelopment of the left side of the heart [1]. CHD represents the most common type of congenital malformation globally, affecting approximately 1% of all live births [2, 3]. Advances in early diagnosis, treatment, and postoperative care have increased the survival of CHD patients, resulting in a steadily growing population of children and adults with CHD [2]. However, studies have demonstrated that even after surgical correction, patients with CHD often continue to face significant challenges into adulthood, including declining cardiac function, reduced exercise capacity, and diminished quality of life [4, 5]. Meanwhile, patients with CHD face not only physiological challenges but also significant psychological issues. Literature indicates that the prevalence of depression among individuals with CHD is substantially higher compared to the general population [6]. Therefore, there is a compelling need for additional interventions, such as cardiac rehabilitation (CR), to address the lifelong management of CHD.

CR is a comprehensive intervention that encompasses various therapeutic modalities, including physical exercise, education on risk factors, behavior modification, and psychological support [7]. The effectiveness of CR in reducing cardiac morbidity and improving health-related quality of life(HRQoL) has been demonstrated in numerous studies [8, 9]. A cardiac rehabilitation program targeting patients with congenital heart disease aged 7 to 24 years demonstrated significant improvements in physical performance following rehabilitation. In addition to enhanced exercise capacity, parent-reported outcomes indicated notable improvements in emotional functioning, social skills, school adaptation, psychosocial well-being, cognitive abilities, and communication [10]. These findings highlight the multidimensional benefits of cardiac rehabilitation in promoting both physical and psychosocial development in CHD patients. Although a substantially increasing number of publications in the field of CR for CHD (CRfCHD) have been published over the two decades [11, 12, 13], there is still a lack of a systematic analysis and summary of this accumulated literature.

Bibliometrics, first proposed by Alan Pritchard in 1969, can analyze a large number of documents in a designated field of study by using mathematical and statistical methods [14]. This approach enables researchers to efficiently comprehend the current research landscape, identify research hotspots, and predict future trends [15]. Therefore, bibliometric analysis has been employed by many researchers to study various domains.

Several bibliometric analyses have been conducted on CR. These studies highlight the increasing attention CR has received due to its proven effectiveness in reducing morbidity and mortality, with a growing body of literature analyzed through tools like CiteSpace to map research trends, interdisciplinary integration, and key contributors. For example, one study analyzed publications from 2001 to 2020, revealing that CR is a multidisciplinary field involving clinical medicine, public health, and sports science, with research hotspots centered on patient education, exercise training, psychological support, and mobile health technologies [16]. Another analysis, focusing specifically on coronary heart disease, emphasized the critical role of physical activity in cardiac rehabilitation, showing steady publication growth and a shift toward research themes such as body composition, participation, and function [17]. However, despite these efforts, there remains a lack of comprehensive bibliometric studies specifically targeting cardiac rehabilitation in patients with CHD, representing a significant gap in the literature that warrants further investigation.

Therefore, this study aimed to provide a comprehensive overview of the existing literature on cardiac rehabilitation for CRfCHD through bibliometric analysis. By examining publication trends, key contributors, research hotspots, and collaborative networks, we sought to identify the current state of research, highlight major developments, and uncover gaps in knowledge. This analysis also serves to inform future research directions and promote the development of more effective and accessible CR programs tailored to the unique needs of individuals with CHD.

Materials and methods

Data collection

In consideration of the diminished relevance of antiquated literature in predicting contemporary hotspots

and trends, we opted to confine the study to publications within the last two decades. Considering the inconsistencies in retrieval strategies and data processing across different bibliographic databases [18], The Web of Science Core Collection (WOSCC) was utilized to gather and extract data on the topic of CRfCHD from January 1st, 2003, to December 31st, 2022, without language restrictions. The search was conducted using the "All Fields" option, ensuring the inclusion of titles, abstracts, author keywords, and other relevant metadata. The search formula TS = (congenital heart disease OR congenital heart diseases OR congenital heart defects OR congenital cardiac OR CHD OR cardiac defect congenital) AND TS = (cardiac rehabilitation OR heart rehabilitation OR rehabilitation OR heart recovery OR cardiac rehabilitation nursing OR cardiac rehabilitation program OR cardiac fitness OR exercise therapy OR physiotherapy OR physical treatment OR psychotherapy) NOT TS = (coronary heart disease) was used, resulting in the collection of 1802 papers.

To ensure the literature's relevance to CRfCHD, the 1802 papers were screened by two independent researchers based on the following inclusion criteria: (1) the literature's title pertains to CRfCHD or its synonyms; (2) the literature's background is related to CRfCHD; (3) subjects are patients with CHD; and (4) literature's purpose and conclusion relate to CRfCHD's mechanism, efficacy, and methods. Papers were excluded if: (1) their main content and central idea had little association with CRfCHD; (2) they were meta-analyses, proceeding papers, letters, meeting abstracts, or book chapters; and (3) their complete information for analyzing could not be provided by WOSCC. In the course of the literature compilation, it was observed that the existing body of literature encompassed individuals across various age groups, including both children and adult populations with preexisting heart disease. Based on the WoSCC web-based personal library, manually excluded articles were removed from the collection accordingly. Upon reviewing literature specifically pertinent to CRfCHD, no explicit delineation of distinctions between children and adult populations was identified. Therefore, a holistic analytical approach encompassing both age groups was adopted. Subsequently, the obtained data is de-duplicated by CiteSpace (Figure S1).

Data analysis

Data was collected from WOSCC, managed and visually analyzed with Microsoft Office Excel 2021 for annual growth trends, CiteSpace, and VOSviewer for institutions, authors, core journals, keywords, and timelines. The free version (basic) of CiteSpace was downloaded from its official website (https://citespace.podia.com/), while the VOSviewer was provided publicly and freely by the Leiden University's Centre for Science and Technology Studies (https://www.vosviewer.com/).

It is worth mentioning that, in CiteSpace, centrality is frequently used to assess the importance of a node. Centrality refers to a measure of prominence or influence assigned to a node. It helps identify the significance of a node in terms of its connections, impact, and position relative to other nodes in the network.

Results

A total of 1,802 articles were initially retrieved from the Web of Science Core Collection (WoSCC). Following the application of predefined inclusion and exclusion criteria, 218 eligible articles were identified. These selected records were exported from WoSCC in plain text format (file name: download_txt) and subsequently used for bibliometric analysis.

The annual growth trend of publications

In this study, an annual analysis was performed to determine the number of research papers that centered on CRfCHD. The results indicated a rising trend in the number of published papers over time (Fig. 1). Specifically, in 2003, there were only 2 published articles involved in

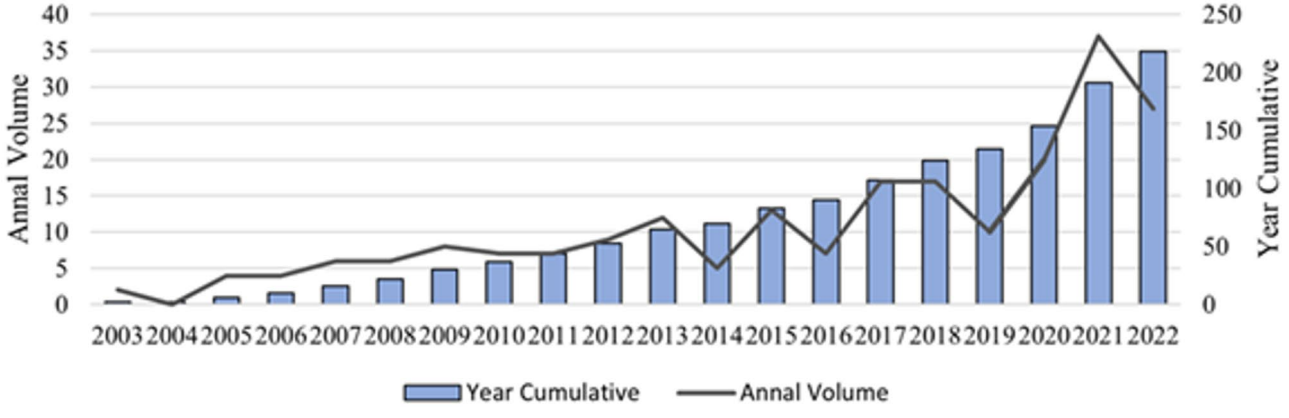

Fig. 1 Publication trends over time

CRfCHD. From 2004 (cumulative volume = 2) to 2013 (cumulative volume = 65), CRfCHD maintained a steady level of attention, with only slight fluctuations observed in the number of related articles, which increased by an average of 7 articles a year. From 2014 (cumulative volume = 70) to 2018 (cumulative volume = 124), the number of published articles on CRfCHD exhibited an increasing trend, with an average annual increase of 13.5 articles. This trend indicates a growing academic interest in this topic during the observed period. Since 2019 (cumulative volume = 134), there has been a significant increase in the number of published articles on CRfCHD, and the cumulative volume reached 218 by 2022, with a growth rate of 28 articles a year, indicating a clear upward trend.

Distribution of countries/regions and institutions

A total of 218 articles from 49 countries and 255 institutions were coauthored by researchers. It is noteworthy that some articles may be affiliated with multiple countries or institutions, given that collaborative research endeavors frequently involve contributions from scholars representing diverse national or institutional affiliations. The United States of America (63) has the largest number of publications, with Canada (36) in second place, followed by Germany (26), the United Kingdom (17), and other countries (Table 1 and Figure S2). Inconsistently, Italy (0.29), Germany (0.16), and Canada (0.15) have higher centrality among the top ten countries, while the United Kingdom (0.13), the United of States (0.11) and Belgium (0.11) are in the second tier (Table 1). The institution with the most articles was University of Toronto (21), followed by Harvard University (19), Boston Children's Hospital (19), Hospital for Sick Children (SickKids) (19), Cincinnati Children's Hospital Medical Center (11), and others (Table 1).

Figure 2 illustrates the nodes representing institutions and the lines between them, indicating their cooperative relationships. The degree and time of cooperation are respectively indicated by the thickness and color of the lines. Over the past two decades, many collaborative studies have been carried out between institutions, but most of the institutions still lack stable and close cooperation with others.

Authors and co-cited authors

Totally 434 authors contributed to the publication of 218 articles included in the analysis, with Mccrindle, Brian W. (7) leading the pack, followed by Ewert, Peter (5), Amedro, Pascal (5), and others (Table 2; Fig. 3). Despite the presence of prolific authors, the study revealed a lack of centrality and cohesive academic networks within the field of CRfCHD.

Co-citation analysis revealed that 33 of the 365 cited authors from the 291 analyzed objects were cited more than 10 times. Longmuir PE (58) was the most frequently co-cited author, followed by Rhodes J (45), Fredriksen PM (37), Takken T (33), Diller GP (31), and others (Table 2). This indicates that these authors are highly influential within the field of CRfCHD research and may have a significant impact on the advancement of the field.

Journals and co-cited journals

The VOSviewer (version 1.6.18) software was utilized to conduct a comprehensive analysis of published journals and co-cited journals. The 218 papers included in this analysis were published in 110 different journals, indicating a diverse range of sources that were considered in this study. Therefore, the journal Cardiology in the Young (24, 10.96%) was found to have the largest number of publications (Table 3). Additionally, only a few journals, namely Cardiology in the Young (24), International Journal of Cardiology (13), Congenital Heart Disease (12), and Pediatric Cardiology (12), had published more than 10 articles each.

To further evaluate the impact of these journals, the 2021 Journal Citation Reports (JCR) were consulted to

Table 1 The top 10 countries/regions and institutions related to CRfCHD

Rank	Country/Region	Average Year	Centrality	Count	Institution	Average Year	Centrality	Count
1	United States of America	2005	0.11	64	University of Toronto	2007	0.07	18
2	Canada	2007	0.15	36	Harvard University	2007	0.03	12
3	Germany	2005	0.16	26	Boston Children's Hospital	2013	0.01	9
4	United Kingdom	2003	0.13	19	Hospital for Sick Children (SickKids)	2005	0.02	8
5	China	2015	0	17	Cincinnati Children's Hospital Medical Center	2017	0.08	8
6	Italy	2006	0.29	13	German Heart Centre Munich	2005	0.04	6
7	Netherlands	2005	0.10	10	Harvard Medical School	2018	0.01	6
8	Australia	2012	0.09	10	University Health Network Toronto	2010	0.04	6
9	France	2011	0.06	10	Academic Medical Center Amsterdam	2017	0	6
10	Sweden	2005	0.06	10	Childrens Hospital of Philadelphia	2014	0.01	5

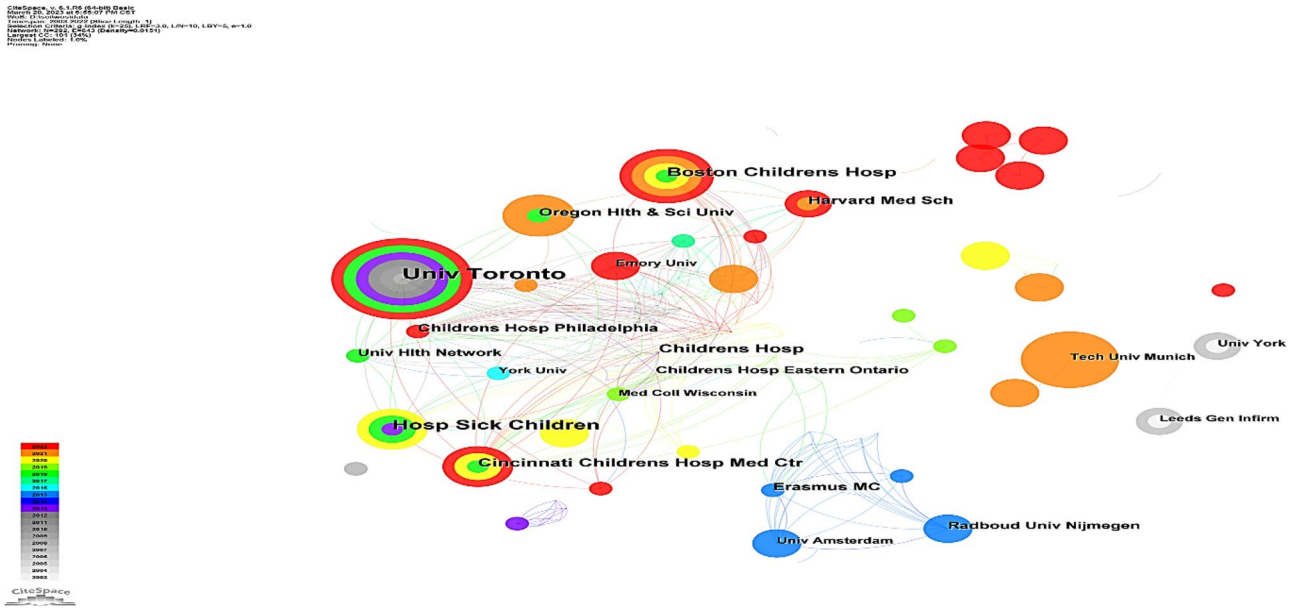

Fig. 2 Institutional cooperation map

Table 2 Top 10 authors and co-cited authors

Rank	Authors	Average Year	Centrality	Count	Co-cited authors	Average Year	Centrality	Count
1	Mccrindle, Brian W	2007	0.02	7	Longmuir PE	2003	0.25	58
2	Ewert, Peter	2018	0	5	Rhodes J	2005	0.08	45
3	Amedro, Pascal	2011	0	5	Fredriksen PM	2003	0.16	37
4	Grace, Sherry L	2009	0.01	4	Takken T	2013	0.04	33
5	Dulfer, Karolijn	2014	0	4	Diller GP	2010	0.1	31
6	Kovacs, Adrienne H	2016	0	4	Moons P	2007	0.1	30
7	Brudy, Leon	2018	0	4	Mccrindle BW	2007	0.04	29
8	Duppen, Nienke	2014	0	3	Duppen N	2015	0.01	28
9	Bredy, Charlene	2017	0	3	Marino BS	2009	0.13	27
10	Banks, Laura	2012	0	3	Warnes CA	2011	0.03	24

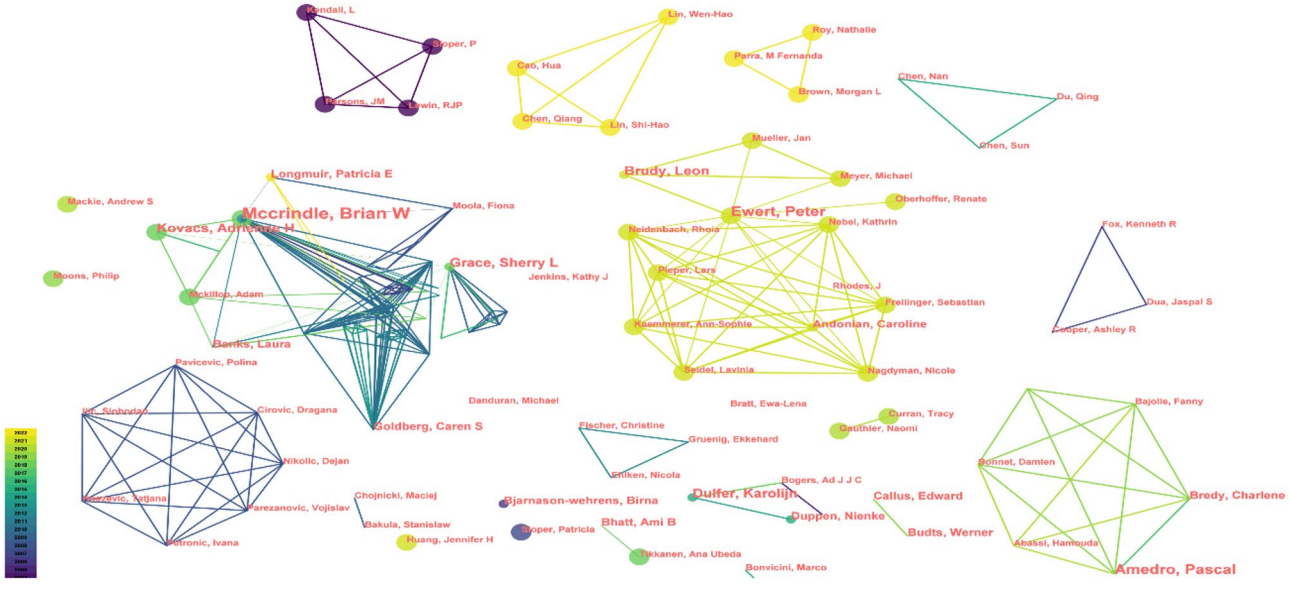

Fig. 3 Author collaboration map

Table 3 Top 10 journals and co-cited journals related to CRfCHD

Rank	Journals	Average Year	Count	IF (2021)	JCR (2021)	Co-cited journals	Average Year	Citation	IF	JCR
1	Cardiology in the Young	2014.88	24	1.023	Q4	Circulation	2005	164	39.922	Q1
2	International Journal of Cardiology	2016.08	13	4.039	Q2	Journal of the American College of Cardiology	2005	127	27.206	Q1
3	Congenital Heart Disease	2016.08	12	2.419	Q3	Cardiology in the Young	2003	114	1.023	Q4
4	Pediatric Cardiology	2016.50	12	1.838	Q4	Pediatrics	2003	110	9.703	Q1
5	International Journal of Environmental Research and Public Health	2020.80	5	4.614	Q2	European Heart Journal	2003	101	35.855	Q1
6	Annals of Thoracic Surgery	2020.50	4	5.113	Q2	International Journal of Cardiology	2006	101	4.039	Q2
7	Journal of the American Heart Association	2020.25	4	6.107	Q2	Heart	2003	99	7.369	Q1
8	Pediatrics	2009.25	4	9.703	Q1	Pediatric Cardiology	2005	94	1.838	Q4
9	Canadian Journal of Cardiology	2020.00	3	6.617	Q2	Congenital Heart Disease	2011	77	2.419	Q3
10	Cardiovascular Diagnosis and Therapy	2020.67	3	2.552	Q3	American Journal of Cardiology	2005	75	3.133	Q3

Table 4 Top 10 co-cited references related to CRfCHD

Rank	Title	Citation	Year	Centrality	References
1	*Systematic review of the effects of physical exercise training programmes in children and young adults with congenital heart disease*	11	2013	0.24	[28]
2	*Promotion of Physical Activity for Children and Adults With Congenital Heart Disease A Scientific Statement From the American Heart Association*	10	2013	0.1	[29]
3	*Impact of cardiac rehabilitation on the exercise function of children with serious congenital heart disease*	9	2005	0.05	[30]
4	*Physical Activity and Sedentary Behavior in Children With Congenital Heart Disease*	9	2017	0.01	[31]
5	*Recommendations for physical activity, recreation sport, and exercise training in paediatric patients with congenital heart disease: a report from the Exercise, Basic & Translational Research Section of the European Association of Cardiovascular Prevention and Rehabilitation, the European Congenital Heart and Lung Exercise Group, and the Association for European Paediatric Cardiology*	8	2012	0.04	[32]
6	*Correlation between cardio-pulmonary exercise test variables and health-related quality of life among children with congenital heart diseases*	6	2016	0.18	[33]
7	*Exercise training in adults with congenital heart disease: Feasibility and benefits*	6	2010	0.13	[34]
8	*Recommendations for participation in competitive and leisure sports in patients with congenital heart disease: a consensus document*	6	2006	0.11	[35]
9	*Exercise training improves activity in adolescents with congenital heart disease*	6	2013	0.04	[36]
10	*Physical activity is associated with improved aerobic exercise capacity over time in adults with congenital heart disease*	5	2013	0.1	[37]

assess their ranking in terms of impact factor and quartile distribution. The top ten journals were found to be distributed among Q1 (10%), Q2 (50%), Q3 (20%), and Q4 (20%). Notably, Pediatrics had the highest impact factor (IF) among these journals, with a score of 9.703.

Citation frequency was also evaluated to assess the influence of these journals within the broader academic community. Of the 430 journals that were cited, only six journals received more than 100 citations (Table 3). Circulation (164) was the most frequently co-cited journal, followed by the Journal of the American College of Cardiology (127), Cardiology in the Young (114), Pediatrics (110), European Heart Journal (101), International Journal of Cardiology (101), and others.

Co-cited references and references burst

CiteSpace was used to comprehensively examine co-cited references in the field of CRfCHD. The results of the analysis are presented in Table 4, which reveals that the article entitled "Systematic review of the effects of physical exercise training programs in children and young adults with congenital heart disease" (11) obtained the highest number of citations and the best centrality score (0.24) among the top 10 cited papers.

The concept of "reference burst" refers to a phenomenon in which the number of citations for a given reference suddenly and rapidly increases. The visualization map of references with citation bursts generated by CiteSpace (Figure S3) illustrates this concept, while the

publication timelines of corresponding articles were represented by blue bars and the timelines of citation bursts were represented by red ones. The top fifty references with the strongest citation bursts primarily began in 2003. Among them, the reference with the highest burst strength was written by Rhodes J. and published in 2005, with a burst strength of 4.77. It's worth noting that the burst of 29 references (58%) occurred in the last decade, and 7 references (14%) were still in the state of citation burst in 2022. These findings suggest that CRfCHD is attracting sustained attention and has the potential for explosive development in the future.

Keywords co-occurrence and hotspots analysis

By examining the related keyword co-occurrence, it is possible to uncover research hotspots and potential directions in specific fields. The 218 articles provided 378 different keywords, and the twenty most common keywords are shown in Table 5. In research involved in CRfCHD, the most common keyword was congenital heart disease (105), followed by physical activity (61), children (60), adolescent (54), quality of life (51), cardiac rehabilitation (48), etc. These topics may be hotspots for CRfCHD-related research. To get a more visual view of hotspots in the CRfCHD domain, a keyword density map was created by VOSviewer (Figure S4). Each point on the map is filled with a color based on the density of the elements surrounding that point; the higher the density, the closer it is to red; conversely, the lower the density, the closer it is to blue. The density size depends on the number of elements in the surrounding area and the importance of those elements.

Clustering keywords on the node graph is an effective approach to investigating the underlying relationships between keywords. Therefore, VOSviewer was used to do it (minimum number of occurrences of keywords ≥ 5). Figure 4 presents a graphical representation of this analysis, in which each node signifies a keyword, with colors indicating distinct clusters. Specifically, all the keywords were assigned to one of six clusters, each reflecting a unique and coherent research direction or area of focus. These clusters were designated as cluster 1 (red), cluster 2 (green), cluster 3 (blue), cluster 4 (yellow), and cluster 5 (purple). The red cluster included congenital heart disease, children, outcome, and care. In contrast, the green cluster contained rehabilitation, exercise capacity, and management. The themes of the blue cluster were exercise, and recommendations, while the yellow cluster primarily focuses on health, young-adults, anxiety, and quality of life. The purple cluster comprised adolescent, cardiac rehabilitation, youth, physical activity, and performance. These findings can aid in understanding the interrelatedness of diverse research themes, ultimately facilitating the design of future studies with more comprehensive coverage of the different facets of the research field.

A timeline viewer merges keywords with time, which enables the display of the keyword evolution process and the developmental path of research hotspots in the entire field. The timeline viewer for these keywords was constructed using VOSviewer (Figure S5). In this figure, the frame on the timeline reflects the average year that the keyword appears. The line between two points means these two keywords appeared in the same article, while the color of the line indicates the published year of it. It is obvious that most of the keywords are located in 2014–2020 after averaging. From 2014 to 2016, most articles on cardiac rehabilitation were closely related to surgery. From 2016 to 2018, there has been a shift in the research emphasis towards exploring the impact of congenital heart disease on various aspects of life, including quality of life among adolescents and children, exercise, anxiety, and failure of life (Figure S6). The keyword that emerged in 2020 signifies a new research direction, with "association" being the most noteworthy among them. This suggests that researchers are placing greater emphasis on the role of collaboration in cardiac rehabilitation.

Discussion

Developing trend and regional disparity

The annual distribution of the articles, as a powerful reflection of the development trend in the field of

Table 5 The top 20 keywords related to CRfCHD

Rank	Keywords	Count	Centrality	Rank	Keywords	Count	Centrality
1	congenital heart disease	105	0.13	11	exercise	21	0.06
2	physical activity	61	0.13	12	health	21	0.09
3	children	60	0.09	13	failure	19	0.09
4	adolescent	54	0.21	14	capacity	17	0.06
5	quality of life	51	0.13	15	outcome	16	0.12
6	cardiac rehabilitation	48	0.21	16	care	14	0.1
7	adult	34	0.15	17	young adult	14	0.03
8	rehabilitation	30	0.09	18	recommendation	13	0.02
9	exercise capacity	25	0.08	19	adult congenital heart disease	13	0.11
10	congenital heart defect	22	0.2	20	disease	13	0.13

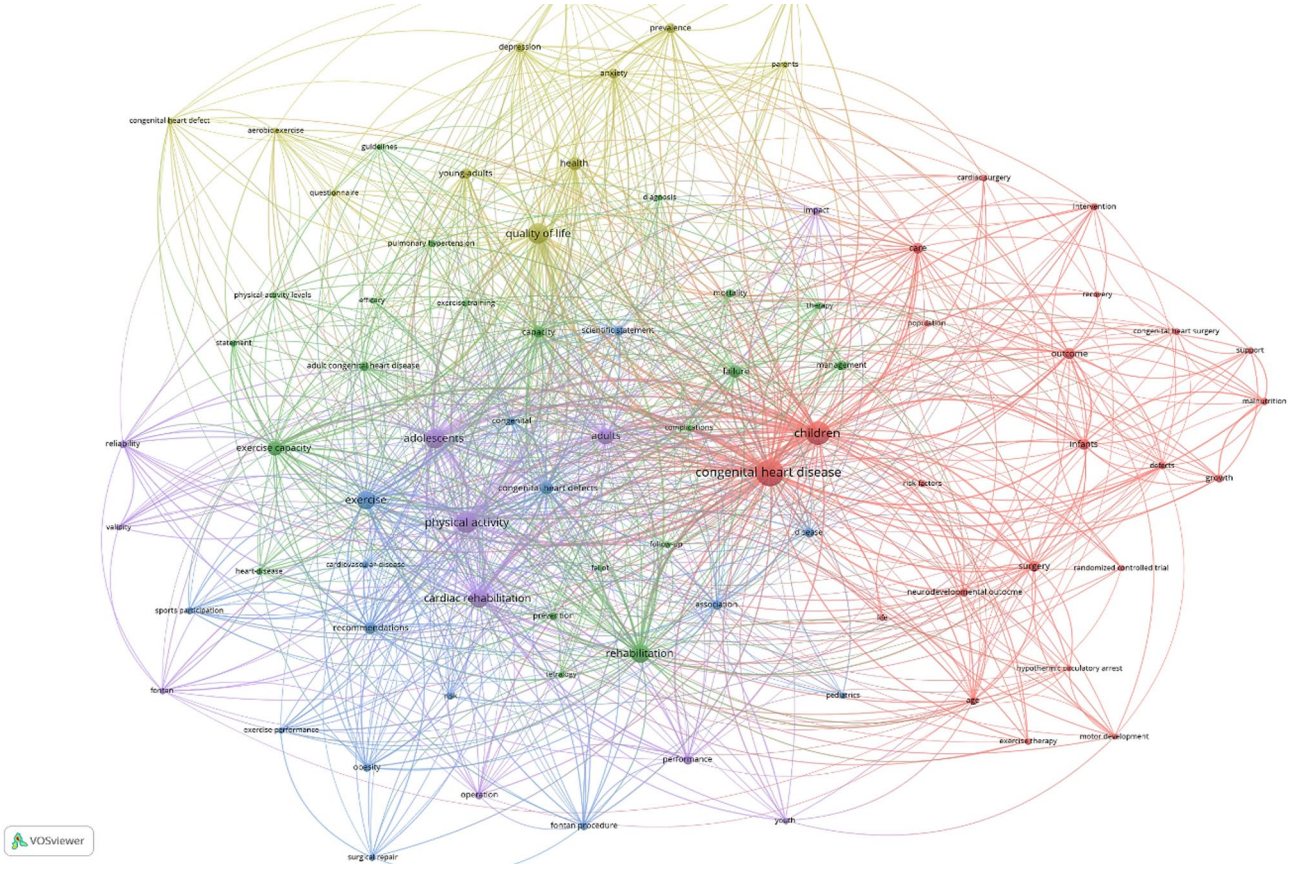

Fig. 4 Keywords network visualization

CRfCHD research, can vividly illustrate the shifting sands of scholarly endeavors. Figure 1 shows a significant increase in the number of published articles since 2019. As shown in Table 1, the USA has the highest number of published papers and the highest centrality in the field of CR. Five of the top 10 organizations in the number of cardiac rehabilitation publications are from the United States and are widely distributed, reflecting the current mammoth growth of cardiac rehabilitation for prevalent heart disease in the United States. Relatively speaking, the University of Toronto contributed nearly half of Canada's total publications in this field, making it the most prolific institution with a comparatively high centrality score. This suggests that while the United States leads in total output, there may be a lack of deep inter-institutional collaboration, as research efforts are more widely dispersed across numerous institutions. In contrast, Canada's publication landscape appears to be more collaboration-oriented; however, current research on CRfCHD remains largely concentrated within the University of Toronto, with limited engagement from other domestic institutions. Meanwhile, only China represents a middle-income country in the top ten countries in terms of publication volume, and no institutions from low-/lower-middle-income countries (LLMICs) are featured among the top ten institutions (Table 1). These findings highlight a pronounced disparity in the focus on CRfCHD among countries classified by the World Bank according to income levels. The disparity in CRfCHD research engagement and cardiac rehabilitation capacity between high-income and low-/lower-middle-income countries is multifactorial, rooted in economic, infrastructural, and workforce limitations. In high-income countries, established healthcare systems, robust research infrastructure, and consistent funding enable comprehensive clinical care and research. Additionally, robust CR programs with structured follow-up care are broadly implemented, and supported by multidisciplinary teams and sufficient funding. By comparison, LLMICs often contend with constrained healthcare budgets, limited registry systems, and inadequate training for specialized professionals, hindering both service delivery and research implementation [19, 20, 21]. Few programs are available in LMICs [22]. Moreover, the financial burden of congenital heart disease in LLMICs is disproportionately severe. For instance, in Niger, the GDP allocated per CHD case is approximately US $0.85, compared to US $525 in high-income settings like Singapore [23]. Additionally, the cost of a single open-heart surgery in Africa and India ranges from US $5,000–10,000—significant

outlays for families in low-resource environments [23, 24].

Given the increasing prominence of CRfCHD research, it is crucial to develop and implement strategies that actively promote the participation of low- and middle-income countries. Concurrently, improving the quality of life for patients with CHD in these regions demands dedicated and focused research efforts. Researchers should prioritize designing tailored approaches to engage countries within these income brackets effectively. Such initiatives are essential to establishing a comprehensive and globally inclusive framework for CRfCHD research and clinical practice, thereby substantially advancing knowledge and enhancing management outcomes in this vital healthcare area.

Journals distribution and knowledge base

As seen in Table 3, only Cardiology in the Young, International Journal of Cardiology, Congenital Heart Disease and Pediatric Cardiology had published more than 10 articles each. This indicated that the field of CRfCHD research is relatively small and specialized, as evidenced by the limited number of publications in major cardiology journals. Further research and development in this field are warranted to address the unique challenges and complexities associated with congenital heart disease in children. Meanwhile, the top ten journals exhibit a distribution across Journal Citation JCR quartiles, with 10% in Q1, 50% in Q2, and 20% each in Q3 and Q4. A noteworthy observation is the relatively sparse representation of articles in Q1 journals, indicating a scarcity of sufficiently authoritative contributions to the field of CRfCHD. As a niche and emerging field, CRfCHD research often lacks the large-scale, high-quality evidence typically favored by top-tier journals. Many studies are observational or involve small sample sizes [25, 26, 27], partly due to the ethical and logistical challenges of conducting research in pediatric populations. This suggests that future CRfCHD studies should aim to increase sample sizes to enhance the scientific rigor and credibility of their findings, thereby improving the quality and impact of published articles.

Co-citation analyses are widely recognized as providing a strong literature basis for the related field. The top ten most co-cited references within the knowledge base (Table 4) were published before 2017, mostly between 2010 and 2013. The most frequently cited reference used a systematic search of the Medline database to identify relevant articles on physical exercise training programs for children and young adults with CHD. It found that most studies reported a significant positive change in the main outcome measure after 12 weeks of regular exercise with no negative findings [28]. The second-most-cited reference highlighted the importance of promoting physical activity in patients with CHD and the need for assessment of individual activity levels, tolerance to exercise, and exercise testing before recommending a particular physical activity [29]. The third reference examined the effects of a cardiac rehabilitation program on exercise performance in children with CHD and identified the physiologic mechanisms responsible for any improvements observed. Its results suggested that routine use of formal cardiac rehabilitation may help reduce the morbidity associated with complex CHD cases in children [30]. The rest of the top ten most co-cited references relate to the promotion of physical activity and the improvement of HRQoL in individuals with CHD. These articles recommend counseling and exercise programs based on specific lesions and complications associated with the disease. These studies found that simple physical activity interventions, motivational techniques, and structured exercise programs can increase physical activity and improve exercise capacity and quality of life in children and adults with CHD [31, 32, 33, 34, 35, 36, 37]. Exercise-based rehabilitation, as one of the core interventions, not only improves cardiopulmonary function but also has a positive impact on mental health. Personalized home-based exercise programs have been shown to significantly enhance children's physical fitness and development, while also improving parents' quality of life by reducing anxiety and caregiving burden [38, 39]. Although some studies suggest that exercise capacity is not always directly correlated with psychosocial quality of life, exercise training has demonstrated beneficial effects on emotional and behavioral issues [40]. This highlights the importance of integrating psychological assessment into exercise rehabilitation programs.

Overall, the top ten most co-cited references focused on summarizing the effects and benefits of physical exercise training programs for children and young adults with CHD. The benefits include improved outcomes in peak VO2 levels, activity levels, and muscle strength, as well as promoting safe engagement in physical activity. These claims were further supported by the original articles through various randomized controlled trials. What's interesting is that among the 10 most co-cited papers, 5 related to pediatric CHD patients, 3 related both children, and adults, while only 2 related to adults, indicating CRfCHD of children and adolescents gained more attention. That is consistent with the relatively higher goal of CR for children with CHD across a range of anatomic variants, which is to help them rise towards their optimal cardiac functioning and enable them to learn to thrive physically, socially, and emotionally [29]. Although there is a wealth of studies on CR in both children and adults with CHD, there is a noticeable gap in understanding the specific differences in CR practices and outcomes between these age groups. This knowledge

gap emphasizes the need for further research that delves into the unique challenges and considerations associated with CR in children and adult populations with CHD. We believe that acknowledging this gap in understanding strengthens the overall contribution of our work by pointing to a crucial area for future investigation.

The analysis of hotspots and frontiers

Keywords are crucial in bibliometrics as they provide insight into the research field's hotspots and trends through techniques such as keyword co-occurrence analysis, keyword cluster analysis, and timeline views. The top twenty keywords in Table 5 reveal the research hotspots in the CRfCHD field, with congenital heart disease (105) being the most representative, followed by physical activity (61), children (60), adolescent (54), quality of life (51), cardiac rehabilitation (48), etc. Again, children and adolescents had higher frequency than adults. Whereas, the most important feature of keyword clustering analysis is to reveal research directions and knowledge structure. In this study, all keywords were clustered into five clusters (Fig. 4), including recommendations for CRfCHD [41, 42], promotion of exercise [43], quality of life before and after CR in patients with CHD [44, 45], the role of CR for different types of CHD [46, 47], and the impact, treatment, and prognosis of CHD [48]. Overall, the field of CRfCHD has been dominated by research on exercise training and quality of life in the last two decades. Numerous studies have focused on the effectiveness of exercise interventions in enhancing the quality of life and exercise capacity of individuals with CHD. Despite these advancements, it is imperative to recognize the complex and multifaceted nature of CHD, which often involves not only physiological impairments but also psychological distress [49]. Unfortunately, there were still few articles related to psychological interventions, indicating the scientific investigation of psychological disorders in CHD patients has been insufficient in comparison to the attention given to physical rehabilitation.

The timeline view of keywords (Figure S6) offers a comprehensive perspective of the evolution of research hotspots, enabling the identification of major research themes and their temporal progression. Before 2016, research on CRfCHD mainly focused on its adjunctive role in surgery, but after 2016, the focus shifted to the role of CR itself. The recent emergence of keywords like association and anxiety suggests that new research directions in CRfCHD involve closer cooperation among various disciplines and psychological interventions for CHD patients.

Citation burst analysis is another common way in bibliometrics to explore abrupt change points and active research frontiers. It's worth noting that three of the seven references still in the citation burst state were authored by Pascal Amedro, which may signify the significant influence of this author in the field. Meanwhile, two of these seven papers involved families, and the other two involved quality of life, suggesting that home-based CR and quality of life may be focuses of current and even future research on CRfCHD.

Based on the above analyses, it is evident that the research hotspots in the field of CRfCHD are centered on exercise training and the improvement of exercise capacity in patients with CHD. In contrast to the emphasis on mortality and complications in the surgical treatment of CHD [50, 51], CRfCHD prioritizes the enhancement of the patient's quality of life. In this regard, family-oriented CR appears to be a promising approach to achieving this goal.

Strengths and limitations

This study utilized the WOSCC database for CRfCHD data, employing CiteSpace and VOSviewer for comprehensive bibliometric analysis. Unlike traditional meta-analysis, bibliometrics excels in visualizing scientific knowledge structures, identifying influencers, and tracking trends. This approach explores research distribution, collaborations, and evolving thematic clusters, offering insights into global landscapes, collaborations, and research foci. Identifying thriving areas, weaknesses, and gaps informs strategic research directions. However, bibliometrics may overlook qualitative nuances and may be biased toward widely disseminated research, excluding unpublished or nonindexed literature.

Conclusion

This study provides a comprehensive summary of the general characteristics of research on cardiac rehabilitation for congenital heart disease (CRfCHD), including contributions by countries, institutions, authors, journals, references, and research directions. Visual analyses using CiteSpace and VOSviewer show a steady upward trend in publications from 2003 to 2022, with the United States and Canada leading the field—particularly the University of Toronto. However, progress in developing countries remains limited, highlighting significant disparities in research capacity and focus. Developing effective CRfCHD protocols that actively involve families and caregivers is essential to improving the quality of life for CHD patients. Future research should place greater emphasis on home-based cardiac rehabilitation and mental health interventions. Additionally, incorporating psychological assessments and tailored interventions within comprehensive CRfCHD programs will be critical. These findings emphasize the need for ongoing research and international collaboration to address challenges and enhance patient outcomes in this specialized field.

Importantly, this study supports the goals of Sustainable Development Goal 3 (SDG-3), which strives to ensure healthy lives and promote well-being for all at every age. By focusing on cardiac rehabilitation for individuals with congenital heart disease—a vulnerable population facing lifelong health challenges—this work contributes to efforts aimed at reducing premature mortality from non-communicable diseases and improving overall quality of life. Strengthening CR programs for CHD patients helps close critical gaps in pediatric and adolescent cardiovascular care, promoting more equitable health outcomes. Moreover, advancing accessible, home-based rehabilitation and integrating mental health services align directly with SDG-3 targets related to universal health coverage and mental health promotion.

Supplementary Information

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Supplementary Material 1

Supplementary Material 2

Author contributions

All authors have contributed to the content of this manuscript and have reviewed and approved it for accuracy and completeness. Conceptualization, D.L.; Data curation, H.Z. and Y.H.; Formal analysis, H.Z. and H.Y.; Funding acquisition, W.Z.; Investigation, H.Z. and Y.H.; Methodology, H.Z. and Y.H.; Project administration, D.L. and W.Z.; Resources, W.Z.; Software, H.Z.; Supervision, D.L. and W.Z.; Validation, H.Z., Y.H. and Z.Y.; Visualization, H.Z.; Writing – original draft, H.Z., Y.H., H.Y. and Z.Y.; Writing – review & editing, D.L. and W.Z. All authors read and approved the final manuscript. All authors read and approved the final manuscript.

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Data availability

Data is provided within the supplementary information files.

Declarations

Competing interests

The authors declare no competing interests.

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